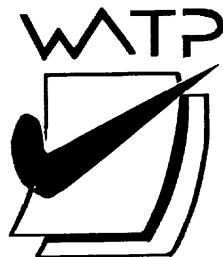


Copyright for test papers and marking guides remains with *West Australian Test Papers*.
Test papers may only be reproduced within the purchasing school according to the advertised Conditions of Sale.
Test papers should be withdrawn after use and stored securely in the school until 4th December.



SEMESTER TWO

MATHEMATICS METHODS UNITS 1 and 2

2020

SOLUTIONS

Calculator-free Solutions

1. (a) $x = 3$ or -3 ✓✓

(b) $f(x) = x^3 + 3x^2 - 9x - 27$

$f'(x) = 3x^2 + 6x - 9 = 0$ ✓

$3(x + 3)(x - 1) = 0$

$x = -3$ or 1 ✓

x		-3		1	
$f(x)$	$\uparrow$	$-$	$\downarrow$	$-$	$\uparrow$
$f'(x)$	$+$	0	$-$	0	$+$

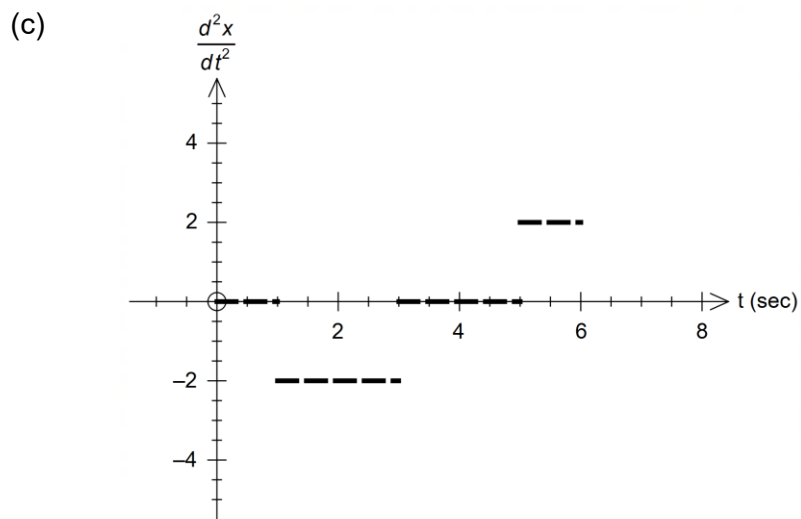
$(-3, 0)$ Maximum ✓

$(1, -32)$ Minimum ✓

(c) $g(x) = f(x + 1) = (x + 1)^3 + 3(x + 1)^2 - 9(x + 1) - 27$ ✓ [8]

2. (a) $0 < t < 1$ or $3 < t < 5$ ✓✓

(b) 6 seconds (< 0) 2 seconds (> 0) ✓



✓✓ [5]

3. (a) (i) $10^{x+5} = 10^8$
 $x = 3$ ✓
- (ii) $x^2 - 1 = 0$ ✓
 $x = \pm 1$ ✓
- (b) $4 \times 48 \times 81 = 2^2 \times 2^4 \times 3 \times 3^4$ ✓
 $= 2^6 \times 3^5$
 $= 2(2 \times 3)^5$ ✓
 $k = 5$ ✓
- (c) 6×10^3 ✓✓ [8]
4. (a) 2π ✓
(b) $(-\pi, -3) (0, 3) (\pi, -3) (2\pi, 3)$ ✓✓
(c) $x = \pi$ or $x = -\pi$ ✓
(d) $-\pi < x < 0$ or $\pi < x < 2\pi$ ✓✓ [6]
5. (a) $2x - 3 - 4x = 4x - 3 - (2x - 3)$ ✓
 $-2x - 3 = 2x$
 $x = -\frac{3}{4}$ ✓
- (b) $T_{n+1} = T_n - 1.5$ $T_1 = -3$ ✓✓ [4]

$$6. \quad (a) \quad m = \frac{\beta^2 - \alpha^2}{\beta - \alpha} = \frac{(\beta + \alpha)(\beta - \alpha)}{\beta - \alpha} \quad \checkmark \checkmark$$

$$\therefore m = \alpha + \beta$$

$$y = (\alpha + \beta)x + b \quad | \quad (\alpha, \alpha^2) \quad \checkmark$$

$$\alpha^2 - \alpha^2 - \alpha\beta = b \quad \checkmark$$

$$\alpha\beta = -b$$

$$(b) \quad (i) \quad y = 2x + 8 \quad \checkmark$$

$$(ii) \quad A(-2, 4) \quad B(4, 16) \quad \checkmark$$

$$d_{AB} = \sqrt{(-2 - 4)^2 + (4 - 16)^2}$$

$$d = \sqrt{180} \quad \checkmark$$

$$= 6\sqrt{5} \text{ units}$$

$$(iii) \quad \frac{-2 + x}{2} = -0.25$$

$$x = 1.5 \quad \checkmark$$

$$y = 1.5^2 = 2.25 \quad (\text{or} \quad \frac{4 + y}{2} = 3.125 \quad y = 2.25) \quad \checkmark \quad [9]$$

7. (a)

	Location A	Location B	Totals
Buses left late	15	40	55
Buses left on time	25	20	45
Totals	40	60	100

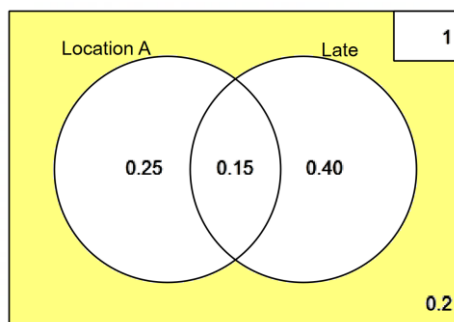
✓✓

(b) (i) The events are whether a bus leaves from Location A or not and whether a bus left late or not. ✓

(ii) Probability of leaving from Location A = 0.4
 Probability of leaving late = 0.55
 $P(\text{Leaving from Location A and leaving late}) = 0.15$ ✓
 $0.4 \times 0.55 = 0.22$ and not 0.15, ✓
 therefore the events are dependent. ✓

(iii) $\frac{0.25}{0.4} = 0.625$ or $\frac{25}{40}$ ✓

(c) (i)



✓✓

(ii) Shading

✓

[10]

Calculator-assumed Solutions

8. (a) $V(10) = 2500$ litres ✓
- (b) 60 mins ✓
- (c) $V'(t) = 2t - 120$ ✓
- $V'(20) = -80$ litres per minute ✓
- (d) $V_2(t) = 7200 \left(1 - \frac{t}{60}\right)^2$
- $V'(t) = 4t - 240$ ✓
- This is $2(2t - 120) \therefore$ double the rate ✓ [6]
9. (a) $T_{n+1} = T_n + 2575$ $T_1 = 52000$ ✓✓
- (b) $A_n = 52000(1.04)^{n-1}$, $n \geq 1$ ✓✓
- (c) \$74 012.21 ✓
- (d) During the 13th year ($83253 > 82900$) ✓✓
- (e) Skye: \$1 548 460.09 ✓
- Indy: \$1 529 250.00 ✓
- Skye's total earnings exceeds Indy's total earnings
- by \$19 210.09 ✓ [10]
10. (a) $V = \pi x^2(2h)$ and $x^2 = R^2 - h^2$ ✓✓
- $V = \pi(R^2 - h^2)(2h)$ ✓
- $\therefore V = 2\pi h(R^2 - h^2)$
- (b) $V'(h) = -6\pi h^2 + 2\pi R^2 = 0$ (for max/min) ✓
- $h^2 = \frac{2\pi R^2}{6\pi} = \frac{R^2}{3}$ ✓
- $h = \frac{\pm\sqrt{3}R}{3}$ Discard negative h
- $\therefore h = \frac{\sqrt{3}R}{3}$ ✓ [6]

11. (a) $T_2 = T_1 + T_3 - 1$ ✓
 $T_2 = x + y - 1$
- (b) $T_3 = T_2 + T_4 - 1$
 $y = x + y - 1 + T_4 - 1$ ✓
 $T_4 = 2 - x$ ✓
Sum = $x + x + y - 1 + y + 2 - x = x + 2y + 1$ ✓
- (c) No common difference nor common ratio ✓✓ [6]
12. (a) F ✓
(b) F ✓
(c) F ✓
(d) T ✓
(e) F ✓ [5]
13. (a) $\lim_{h \rightarrow 0} \frac{(2x + 2h + 3)^2 - (2x + 3)^2}{h}$ ✓✓
- (b) $y = x^2 \therefore \frac{dy}{dx} = 2x$ when $x = 3$ ✓
 $2(3) = 6$ ✓
- (c) $\frac{dy}{dx} = 6x - 2 = 4 \therefore x = 1$ ✓
 $y = 3(1)^2 - 2(1) + 1 = 2 \quad y = 2$ ✓
- (d) $\frac{17}{2}$ ✓ [7]
14. (a) $y = 3 + 2^x$ D ✓
- (b) $y = 2^x$ C ✓
- (c) $y = 2^x - 3$ A ✓
- (d) $y = \left(\frac{1}{2}\right)^x$ B ✓ [4]

15. (a) $l = r\theta$
 $\frac{63\pi}{20} = (2.7)\theta$ ✓
 $\theta = \frac{7\pi}{6}$ $\frac{2\pi}{12} = 1 \text{ hour}$ $\therefore$ this has taken 7 hours. ✓✓
- (b) $A = \frac{1}{2} r^2 \theta = \frac{1}{2} (2.7)^2 \left(\frac{\pi}{3}\right)$ ✓
 $= \frac{243\pi}{200} m^2$ ✓ [5]
16. (a) $t = 0 \therefore x(0) = 0 \text{ m}$ ✓
- (b) $x(t) = t^3 - 9t^2 + 16t$
 $v(t) = 3t^2 - 16t + 16$ ✓
 $v(0) = 16 \text{ m/s}$ ✓
The particle is initially travelling to the right of the origin. ✓
- (c) $(t - 4)(3t - 4) = 0$
 $t = 4$ or $t = \frac{4}{3}$ First changes direction at $\frac{4}{3} \text{ s.}$ ✓
- (d) $a(t) = 6t - 16$ ✓
 $a(3) = 2 \text{ m.s}^{-2}$ ✓
- (e) $v(3) = -5 \text{ m.s}$ ✓
Velocity is negative and acceleration is positive therefore the particle is slowing down. ✓
- (f) First turns at $\frac{4}{3} \text{ s.}$ $x\left(\frac{4}{3}\right) = \frac{256}{27} \text{ m} \text{ (9.48 m)}$ ✓
Turns again at 4 s. $x(4) = 0$ ✓
 $x(5) = 5 \text{ m}$
 $\therefore$ Total distance = $2\left(\frac{256}{27}\right) + 5 = \frac{647}{27} \text{ m} \text{ (or 23.96m)}$ ✓ [12]

17. (a) $2 = r^{12}$
 $r = 1.0595$ ✓
 $\therefore$ 5.95% per hour is the rate of growth
- (b) $9000 = 1000(1.0595)^t$ ✓
 $t = 38.016$
 Approximately 38 hours. ✓
- (c) $P = 9000\left(\frac{2}{3}\right)^8$ ✓
 $P = 351$ bacteria left ✓
 Their expectations were not accurate. The anti-biotic killed more than a third of the bacteria per hour. ✓
OR $9000(r)^8 = 1$
 $r = 0.3204$ The antibiotic killed 68% per hour which is more than $\frac{2}{3}$. [6]

$$18. \quad (a) \quad (i) \quad \frac{1-t}{t+1} = \frac{2-5t}{1-t} = r \quad \checkmark$$

$$6t^2 + t - 1 = 0$$

$$t = \frac{1}{3} \text{ or } -\frac{1}{2} \quad \checkmark$$

$$\text{For } t = \frac{1}{3} \quad r = \frac{1 - \frac{1}{3}}{\frac{1}{3} + 1} = \frac{1}{2} \quad \checkmark$$

$$\text{For } t = -\frac{1}{2} \quad r = \frac{1 + \frac{1}{2}}{-\frac{1}{2} + 1} = 3 \text{ Discard} \quad \checkmark$$

$$\text{Common ratio} = \frac{1}{2} \text{ when } t = \frac{1}{3}$$

$$(ii) \quad S_{\infty} = \frac{a}{1-r} \quad \text{when } a = t + 1 = \frac{4}{3} \quad \checkmark$$

$$= \frac{8}{3} \quad \checkmark$$

$$(b) \quad (i) \quad S_8 = 8^2 - 2(8) = 48 \quad \checkmark$$

$$(ii) \quad S_7 = 7^2 - 14 = 35 \quad \checkmark \checkmark$$

$$T_8 = S_8 - S_7 = 13$$

$$(iii) \quad S_{15} - S_{12} = (15^2 - 30) - (12^2 - 24) \quad \checkmark$$

$$= 75 \quad \checkmark$$

$$(c) \quad T_1 = 27 - 6(1 + 1) = 15$$

$$T_2 = 27 - 6(2 + 1) = 9$$

$$a = 15 \quad d = -6 \quad \checkmark \checkmark$$

$$S_{20} = -840 \quad \checkmark$$

[14]

19. (a) Roots at $(0, 0)$ $(2, 0)$ $(-2, 0)$ ✓

(b) $f'(x) = 4x^3 - 8x = 4x(x^2 - 2) = 0$ (for stationary points) ✓

Stationary points at $(0, 0)$ $(\sqrt{2}, -4)$ $(-\sqrt{2}, -4)$ ✓

$f''(x) = 12x^2 - 8$ $f''(0) < 0 \therefore \text{Max}$ ✓

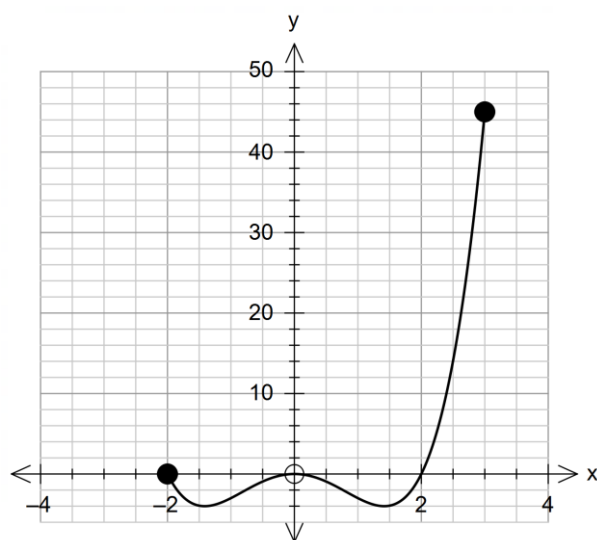
$f''(-\sqrt{2}) > 0 \therefore \text{Min}$ ✓

$f''(\sqrt{2}) > 0 \therefore \text{Min}$ ✓

Or Sign table:

x		$-\sqrt{2}$		0		$\sqrt{2}$	
$f(x)$	$\downarrow$	$-$	$\uparrow$	$-$	$\downarrow$	$-$	$\uparrow$
$f'(x)$	$-$	0 Min	$+$	0 Max	$-$	0 Min	$+$

(c)



✓✓✓

[9]

20. (a) (i) $3x^2 - 6x + 3 = 12$

$$x = -1 \text{ or } 3$$

$$(-1, -3) \quad (3, 13)$$

✓✓

(ii) $y = 3x + 4 \quad m = 3$

$$3x^2 - 6x + 3 = 3$$

$$x = 0 \text{ or } x = 2$$

$$(0, 4) \quad (2, 6)$$

✓✓

(iii) $3x^2 - 6x + 3 = 27$

$$x = -2 \text{ or } x = 4$$

$$(-2, -22) \quad (4, 32)$$

✓✓

(b) $(-2)^2 + a(-2) + b = 0$

$$\therefore 2a - b = 4$$

And $a + b = -1$

$$\therefore a = 1 \text{ and } b = -2$$

✓✓

$$f(x) = \frac{x^3}{3} + \frac{x^2}{2} - 2x + c \quad | (2, 3)$$

✓

$c = \frac{7}{3}$ The equation of the curve is $f(x) = \frac{x^3}{3} + \frac{x^2}{2} - 2x + \frac{7}{3}$. ✓ [10]

End of Questions